

# Mental Health as EM Substrate Stability: A Compiled Framework Paper

A Theoretical and Technical Framework for Electromagnetic Regime Analysis  
in Psychiatric Contexts

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Classification: Compiled Framework Document — Research Draft

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## ABSTRACT

This paper presents a physically grounded framework for describing mental health in terms of electromagnetic (EM) substrate stability. The central claim is that mental states — including both healthy and disordered ones — can be characterized as attractor regimes within a high-dimensional EM feature space generated by the biological activity of the brain and body. The framework covers substrate definition, regime modeling, noise physics, sensor geometry, acquisition pipelines, and clinical mapping. It concludes with an honest account of what current technology can and cannot yet deliver, and what problems must be solved before this framework reaches clinical utility.

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## SECTION 1

### SUBSTRATE DEFINITION

The electromagnetic substrate, as used in this framework, refers to the organized, persistent pattern of EM field activity generated by the collective biological processes of the human body and brain. This includes ionic flows across neuronal membranes, the synchronized firing of neural populations,

glial cell activity, cardiac electrical cycles, and the dynamic polarization and depolarization of cellular membranes throughout the body. The substrate is not a metaphor or an abstraction — it is the aggregate of physically real, measurable electromagnetic fields existing in and around biological tissue at any given moment.

The body produces measurable EM fields across a wide range of frequency bands. At the low end, DC bioelectric potentials arise from standing electrochemical gradients maintained by cellular ion pumps. Extremely low frequency (ELF) fields in the range of 0.1 to 100 Hz correspond closely to neural oscillatory activity measurable via EEG and MEG. Higher frequency components extend into the radiofrequency range, though their biological significance and measurability vary considerably with the technology available. The substrate, then, is not a single signal but the aggregate of all these contributions within a defined spatial region over a defined time window.

It is important to distinguish the substrate from the underlying neural activity that gives rise to it. Neural activity — the electrochemical signaling of individual neurons and ensembles — is a biological process involving matter, membrane dynamics, and chemistry. The EM substrate is the electromagnetic expression of that activity. Once generated, the resulting fields propagate outward from their source tissue and exist partly independent of it. A magnetic field produced by a neural current loop is a physical entity in space; it can be detected at a distance from the tissue that produced it. This is the basis on which MEG and EEG operate.

The substrate can be characterized along several dimensions: its spectral distribution (how field power is distributed across frequency bands), its coherence (the degree to which oscillations at different spatial locations are correlated), its phase stability (how consistently oscillations maintain their temporal relationships), the amplitude envelope of its dominant components,

and its spatial topology (the three-dimensional geometry of field patterns across and around the head or body). Each of these dimensions carries information about the functional state of the underlying biological system.

Taken together, these dimensions define a rich representational space for the substrate's state at any moment. A complete substrate characterization at a single point in time would be a vector in this high-dimensional space — a specific combination of spectral, coherence, phase, and spatial values. Understanding mental health in EM terms begins with accepting that this vector is not merely a correlate of mental state but, within this framework, constitutes the physical substrate from which mental states emerge.

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## SECTION 2

### **REGIME MODEL**

A regime, in the context of this framework, is a stable attractor state within the high-dimensional EM feature space of the substrate. Rather than existing in one fixed configuration, the substrate moves continuously through this space, but it does not wander randomly. Instead, it tends to cluster around certain recognizable configurations — patterns of field organization that persist over timescales longer than individual oscillations. These clusters are the regimes. They represent states the system preferentially occupies and returns to after perturbation, in the same way that a ball in a hilly landscape tends to settle into valleys rather than remain on slopes.

Regimes are defined by characteristic signatures across multiple dimensions simultaneously. A regime is not identified by a single spectral peak or a single coherence value but by a combination: a particular alpha-band power level paired with a particular frontal-parietal coherence ratio, a specific phase-locking pattern between hemispheres, and a characteristic amplitude modulation profile. Different brain states produce distinct and reproducible

regime signatures. Focused, alert wakefulness produces a regime with suppressed alpha, elevated gamma, and strong frontal coherence. Resting eyes-closed states show a different coherence topology and dominant alpha rhythm. Sleep cycles produce a sequence of progressively distinct regimes — from light NREM through slow-wave sleep to REM — each with its own clear EM signature. Acute stress, depressive episodes, manic states, and psychotic states each produce their own characteristic, measurable deviations from the waking resting baseline.

Formally, a regime  $R$  can be defined as a bounded region in a high-dimensional feature space  $F$ , where  $F$  is constructed from spectral power values across defined frequency bands, coherence matrices between measurement sites, and phase-amplitude coupling indices. At any moment, the substrate occupies a point  $P$  in  $F$ . If  $P$  falls within the region  $R$ , the system is in regime  $R$ . Multiple regimes can be defined, and the full regime map of an individual describes which regions of  $F$  they tend to occupy, how long they remain in each, and how likely they are to transition between them.

Transitions between regimes are not always gradual. The system can shift abruptly from one attractor basin to another — a phenomenon familiar in dynamical systems as a bifurcation or critical transition. Near regime boundaries, the system becomes more sensitive to perturbation: small external inputs or internal fluctuations can push the feature vector across a threshold into a different attractor. This non-linearity is significant: it means that the relationship between cause and effect in mental state transitions is not always proportional. A small pharmacological change or a minor psychosocial stressor may be sufficient to trigger a large-scale regime shift when the system is near a boundary, while the same input far from a boundary produces no regime change at all.

The deeper implication of the regime model is that mental states are not purely neurochemical phenomena — they are also topological features of EM

organization. A mental state does not merely correlate with a particular pattern of field activity; within this framework, the field pattern is the physical realization of the state at the substrate level. This does not deny the role of neurochemistry — neurotransmitter concentrations directly influence the ionic dynamics that generate the fields — but it situates neurochemistry as one of several mechanisms that shape regime structure, rather than as the sole level at which mental states are defined.

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## SECTION 3

### **NOISE PHYSICS**

Any attempt to measure the EM substrate must contend with noise — signals that obscure the biological fields of interest. Noise sources in substrate measurement fall into two broad categories: internal biological noise arising from the subject's own body, and external environmental noise from sources outside the body. Both categories must be understood and managed for meaningful regime-level measurement to be possible.

Internal biological noise includes cardiac EM artifact, which produces a strong, rhythmic signal from the QRS complex of the ECG that is often visible in both EEG and MEG recordings. Skeletal muscle activity (EMG) generates broadband high-frequency interference that can overwhelm neural signals, particularly in scalp EEG where electrodes are near facial and scalp muscles. Respiratory motion produces slow artifact due to the movement of the body and the resulting changes in electrode-skin contact and in the position of the head or body relative to the sensors. Skin-electrode interface effects, including impedance fluctuation and electrodermal activity, introduce low-frequency drift and noise in EEG systems. These internal sources are problematic precisely because they are physiologically correlated with the

subject's state, making simple temporal filtering insufficient for their removal without risk of distorting the signals of interest.

External environmental noise presents a different set of challenges. Power-line interference at 50 Hz (in Europe and much of the world) or 60 Hz (in North America) and their harmonics is ubiquitous in any environment with electrical infrastructure. Wireless communication devices produce RF interference across a wide band. Mechanical vibration — from building infrastructure, HVAC systems, or nearby traffic — can couple into sensors through physical contact or through modulation of magnetic field gradients. Geomagnetic fluctuations, though small in magnitude, can affect sensitive magnetometers operating in unshielded environments. The scale mismatch is severe: environmental magnetic fields are typically in the nanotesla range, while the neural signals detectable by MEG are in the femtotesla to picotesla range — a difference of three to six orders of magnitude.

At the sensor level, two fundamental noise sources set a physical floor on measurement sensitivity. Thermal noise, also called Johnson-Nyquist noise, arises from the random thermal motion of charge carriers in resistive elements of the sensor circuit. Its power spectral density is proportional to the sensor resistance and to temperature, which is why some high-sensitivity systems use cryogenically cooled superconducting sensors (SQUIDs) — reducing temperature dramatically reduces this noise floor. Shot noise arises in photodetectors used in optically pumped magnetometer (OPM) systems, where the discrete, quantized nature of photon detection introduces statistical fluctuations in measured photocurrent. Both noise types set a minimum detectable signal level for a given sensor and integration time.

The concept of noise-limited regime resolution captures the practical consequence of all this: the minimum detectable difference between two regime states, given a specific sensor configuration and noise environment. If two regimes differ in their feature-space representation by an amount smaller

than the effective noise in the measurement system, those regimes cannot be distinguished. This is not merely a technical inconvenience — it is a fundamental constraint on the clinical granularity achievable with substrate scanning. Improving regime resolution requires either reducing noise at the sensor and environment level, increasing signal amplitude through better source-sensor coupling, or using longer integration times and statistical aggregation across sessions. Each of these strategies carries its own tradeoffs in terms of cost, practicality, and temporal resolution.

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## SECTION 4

### **SHIELDING AND GEOMETRY**

Magnetically shielded rooms (MSRs) are the standard operating environment for high-sensitivity EM substrate measurement. An MSR is a room constructed with multiple concentric shells of high-permeability metal alloy — typically mu-metal — that redirects external magnetic field lines around the enclosed space rather than through it. High-quality MSRs achieve attenuation of external fields by factors of 10,000 or more at EEG and MEG frequencies. More advanced installations supplement passive shielding with active compensation systems: arrays of coils driven by real-time feedback loops that sense residual fields inside the room and generate opposing fields to cancel them. Together, these systems create an environment in which the neural EM signals of interest are no longer buried beneath environmental interference, making femtotesla-level measurement possible.

Sensor geometry is as important as shielding in determining what can be measured. The placement of sensors relative to the head determines which sources contribute to each measurement and how much spatial information can be extracted. Sensors farther from the head receive a more blurred, mixed signal from a larger volume of tissue. Sensors closer to the surface

receive more spatially specific information, but their coverage is smaller. The challenge of reconstructing source locations from surface measurements — the inverse problem — is inherently underdetermined: infinitely many spatial source configurations inside the head could produce the same pattern of measurements on the outside. Geometry determines how severely ill-posed this problem is, and thus how much uncertainty remains in source localization even after the best available algorithms are applied.

Different sensor configurations represent different tradeoffs in this geometric space. Standard EEG systems using the 10-20 electrode placement scheme deploy 19 to 21 electrodes across the scalp, providing broad coverage but limited spatial resolution. High-density EEG systems with 128 to 256 channels reduce inter-electrode spacing, improving spatial sampling at the cost of increased setup time and technical complexity. Conventional MEG systems use a helmet-shaped array of typically 100 to 300 SQUID sensors positioned around the head but separated from the scalp by the cryogenic dewar housing. Newer OPM-based systems allow sensors to be placed directly on the scalp surface — significantly reducing the sensor-to-source distance and improving the spatial resolution achievable for a given number of sensors. OPM systems also have the advantage of being operable without cryogenic cooling, though they introduce their own noise and motion-sensitivity challenges.

Coverage and resolution stand in tension with each other given finite resources. Increasing the number of sensors improves both spatial coverage and, in principle, spatial resolution, but also increases wiring complexity, the risk of common-mode noise pickup across many channels, computational demands of data processing, and cost. The optimal configuration depends on the clinical or research question: whole-scalp regime classification requires broad coverage, while source-resolved coherence mapping benefits from



dense, closely spaced sensors over specific regions. No single configuration is optimal for all purposes.

An emerging extension of the geometry question is the inclusion of body-level sensors beyond the head. The cardiac system, respiratory system, and peripheral nervous tissue all contribute to the total EM substrate. Whole-body sensor arrays — placing electrode or OPM coverage across the torso, limbs, and head simultaneously — would in principle capture the full substrate, allowing cardiac-neural coupling, peripheral autonomic contributions, and limb-level EM activity to be included in regime characterization. This is technically demanding and not yet standard practice, but it reflects the logical endpoint of treating the substrate as a body-level phenomenon rather than a brain-only one.

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## SECTION 5

### **STRUCTURAL VS. REGIME SCANS**

Structural brain imaging modalities — primarily MRI and CT — produce detailed maps of the physical architecture of the brain. MRI reveals tissue type, volume, gray and white matter boundaries, and, in diffusion-weighted variants, the orientation and integrity of white matter fiber tracts. CT is faster and better suited to detecting bone abnormalities or acute hemorrhage. Both are static: they capture a snapshot of the physical structure of the brain at a single point in time and say nothing directly about the ongoing functional state of the tissue. A structural scan tells you what is there; it does not tell you what it is doing at the moment of acquisition or at any other moment.

Regime scans, by contrast, are inherently dynamic. They capture the EM state of the system across time, characterizing which regimes the substrate occupies, for how long, and in what sequence. A regime scan of a person in a depressive episode, a person in a manic state, and a healthy control may

produce structurally similar brains under MRI while producing clearly distinct regime distributions under MEG or EEG. This is consistent with what is observed clinically in many psychiatric conditions: major depressive disorder, generalized anxiety disorder, and schizophrenia are not reliably associated with gross structural brain abnormalities visible on standard MRI in the majority of patients. The pathology, within this framework, is in the EM dynamics — the regime — not in the tissue architecture.

Despite this distinction, structural and regime data are complementary rather than competing. Structural data provides the spatial prior — the anatomical map that tells us where different tissue types, cortical regions, and white matter pathways are located in a given individual. This information is essential for regime data interpretation: without knowing the anatomy of the brain underlying a set of scalp-surface EM measurements, source reconstruction is severely underconstrained. The standard practice in MEG research is to co-register the MEG data with an individual structural MRI, using the anatomy to guide the source-localization models that translate surface field patterns into estimates of where, inside the brain, the sources are located.

Diffusion tensor imaging (DTI), a structural MRI technique that maps the directional diffusivity of water in white matter to infer axonal connectivity, adds another layer of complementarity. DTI connectivity data reveals which brain regions are structurally connected by fiber pathways and provides an anatomical basis for predicting which regions are likely to show correlated EM activity. If two regions are strongly connected by a dense, myelinated fiber tract, they are expected to show stronger coherence in relevant frequency bands than regions with sparse or absent direct connectivity. DTI-informed priors can thus improve the interpretability and specificity of coherence-based regime analysis by distinguishing coherence arising from

direct structural coupling from coherence arising from common input or volume conduction.

The integrated scan concept — a single session that acquires both structural MRI and dynamic EM (MEG or OPM) data — represents the ideal workflow for this framework. In practice, the technical requirements of the two modalities are often incompatible in the same physical space: MRI requires strong static magnetic fields that would destroy MEG sensors, and the two modalities are typically performed in separate sessions and then co-registered afterward. Work is underway in several research groups to develop hybrid acquisition workflows that minimize the time between structural and regime acquisitions and standardize the co-registration process. Until such integrated workflows are routine and validated, the structural-regime data fusion required for optimal regime interpretation remains a multi-session, multi-step process with associated co-registration error.

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## SECTION 6

### **SIGNAL ACQUISITION PIPELINE**

Stage 1 of the acquisition pipeline is the sensor interface. This involves the physical placement of electrodes, coils, or OPM sensors in the correct positions relative to the head or body. For EEG, electrodes must be applied with conductive gel or saline solution to achieve low impedance contact with the scalp, and impedance is verified at each site before recording begins. A reference electrode — typically placed at a location of low biological signal, such as the mastoid or the linked earlobes — establishes the baseline against which all other measurements are made. For OPM systems, a zero-field reference configuration is used, with at least one sensor placed away from biological signal sources to characterize the ambient field environment.

Proper sensor interface is foundational: errors here — poor contact, incorrect placement, high impedance — propagate through every subsequent stage of the pipeline and cannot be corrected after acquisition.

Stage 2 is analog signal conditioning. The raw signal from each sensor is typically in the microvolt range for EEG and requires differential amplification — amplifying the difference between each recording electrode and the reference, while rejecting common-mode interference that appears identically at both. Hardware bandpass filtering removes DC offset and very slow drift at the low end, and attenuates high-frequency components above the range of interest at the high end, preventing aliasing during digitization. Notch filters centered at the power-line frequency and its harmonics are applied in hardware to suppress the dominant environmental interference before the signal reaches the digitizer. The goal of this stage is to maximize the signal-to-noise ratio of the biologically relevant frequency content entering the digital domain.

Stage 3 is digitization: the conversion of the analog signal to a digital time series. The Nyquist theorem requires a sampling rate of at least twice the highest frequency of interest to avoid aliasing. For broad-band EEG or MEG recordings intended to capture frequencies up to 500 Hz or beyond, sampling rates of 1,000 to 5,000 samples per second are standard. Bit depth determines the dynamic range of the digitized signal — the ratio between the largest and smallest representable value. Modern EEG systems use 24-bit analog-to-digital converters, providing over 16 million discrete levels, which is necessary to represent both the small biological signals and any residual large-amplitude artifacts within the same recording without overflow or loss of resolution at the low end.

Stage 4 is digital preprocessing. Independent component analysis (ICA) is the most widely used technique for separating biological artifacts from neural signal in the digital domain. ICA decomposes the multi-channel recording into

statistically independent components, which can then be individually inspected and those corresponding to cardiac artifact, eye movement, muscle noise, or other known artifact sources can be removed before reconstructing the cleaned signal. Signal-space projection (SSP) is an alternative approach, particularly common in MEG, that identifies artifact spatial patterns from reference data and projects them out of the signal subspace. Epoching — segmenting the continuous time series into defined windows — follows preprocessing and prepares the data for feature extraction.

Stages 5 through 7 constitute the analytical backend. Feature extraction (Stage 5) computes the metrics that define the regime feature space: power spectral density at defined frequency bands, coherence matrices between all pairs of sensor sites, phase-amplitude coupling indices, and related measures. Regime classification (Stage 6) applies a trained model or reference regime map to the extracted feature vector, assigning it to a regime or computing its distance from known regime attractor centers. Output generation (Stage 7) produces the clinical-facing report: regime state label, transition probability matrix, stability score, and derived indicators. Each stage introduces potential error — from ICA misclassification of artifact components to regime model overfitting — and the total uncertainty propagating through the pipeline must be characterized and reported before any output can be meaningfully interpreted in a clinical context.

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## SECTION 7

### **MENTAL HEALTH AS EM STABILITY**

The central theoretical claim of this framework is direct: mental health, defined functionally, is a property of the stability and appropriate flexibility of the EM regime. A mentally healthy system, in these terms, is one that occupies regimes appropriate to its current context — alert and engaged

during demanding tasks, rested and diffuse during recovery, emotionally regulated under social and environmental stress — and transitions fluidly between them when circumstances change. It is not a system locked into any single regime but one that navigates the regime landscape adaptively, entering the right attractor at the right time and exiting it when conditions shift.

This framing maps onto clinical presentations in specific and testable ways. Major depressive disorder, under this model, corresponds to a system trapped in a low-arousal, high-default-mode-network-dominance, low-frontal-coherence regime. The system fails to exit this regime in response to stimuli that would ordinarily trigger a shift toward higher engagement — it is stuck in an attractor basin that is too deep or too wide, with insufficient energy to escape. Anxiety disorders, by contrast, correspond to a system with insufficient regime stability: it transitions too rapidly and without appropriate contextual cause, producing the phenomenological experience of uncontrolled worry, hyperarousal, and difficulty settling. The feature vector moves erratically through the regime space rather than settling into appropriate attractor basins.

Bipolar disorder, within this model, represents a system with an abnormally wide attractor landscape — one in which the system can reach regime states that are far outside the range accessible to a healthy system. During mania, the system occupies a high-arousal, high-connectivity, high-energy regime with reduced frontal regulation. During depression, it occupies the opposite extreme. The transitions between these states may be relatively abrupt, consistent with the bifurcation dynamics discussed in Section 2. Schizophrenia presents a different picture: not necessarily extreme regime positions, but disrupted coherence between regions that are normally tightly coupled, producing regime states that are internally inconsistent — where

the global pattern of EM organization lacks the normal binding and integration of distributed neural activity.

This framework is not a replacement for existing psychiatric nosology or for neurochemical accounts of psychiatric disease. Neurotransmitter systems, receptor pharmacology, genetic predispositions, developmental factors, and psychosocial context all remain relevant. The EM substrate model adds a substrate-level description to these existing accounts — one that is potentially measurable in real time, in individuals, and that bridges the gap between molecular mechanisms and observable behavior. The regime concept is a meso-level description: coarser than the single-neuron level, finer than the symptom-cluster level, and potentially more tractable as a target for measurement and intervention than either extreme.

Two derived concepts are particularly important for clinical application. The first is regime resilience — the speed and reliability with which the system returns to adaptive regimes after perturbation. Regime resilience may be a measurable EM correlate of psychological resilience, potentially detectable as a feature of the transition matrix rather than of the regime positions themselves. The second is the regime-level interpretation of treatment effects. Pharmacotherapy alters the substrate chemistry — shifting receptor kinetics, altering ion channel dynamics — and thereby changes the shape of the regime landscape: moving attractor positions, altering basin depths, or enabling new transition pathways. Psychotherapy, under this model, may operate by retraining the conditional probabilities of regime transitions through repeated experience — a form of EM-level habit formation. Neurostimulation techniques such as TMS and tDCS directly perturb the EM substrate through externally applied fields, potentially destabilizing maladaptive attractor basins or reinforcing adaptive ones, depending on the timing, frequency, and spatial targeting of the stimulation.

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## OMNISCAN OUTPUTS

The OmniScan is a proposed integrated multi-modal EM substrate acquisition session that combines simultaneous EEG or OPM-MEG, ECG, respiratory monitoring, and optional electro-oculography (EOG) and EMG into a single time-locked recording. The rationale for this integration is that the full substrate involves more than just the brain: cardiac, respiratory, and peripheral contributions to the EM environment are relevant both as sources of information and as sources of artifact that must be accounted for. Time-locking all channels to a common acquisition clock allows the relationships between these signals — not just their individual properties — to be analyzed.

The primary outputs of an OmniScan session are as follows. The Regime State Map is a time-series of regime classifications showing which regime the subject occupied at each moment during the session, typically displayed as a sequence of labeled intervals or as a trajectory through a two- or three-dimensional projection of the regime feature space. The Transition Matrix is a probabilistic description of how frequently and how readily the subject moved between each pair of regimes during the session — the  $ij$ -th entry of the matrix gives the probability that, given the subject was in regime  $i$ , they transitioned to regime  $j$  in the next time step. The Stability Score is a scalar summary, derived from the variance of the feature vector within each occupied regime, indicating how tightly the subject's substrate clustered around the attractor centers of the regimes it occupied.

The Coherence Topology Map is a spatial representation of inter-site coherence at specific frequency bands, computed from the cleaned multi-channel recording and projected onto a template or individual head model. It visualizes which pairs of regions were coherently coupled at which frequencies, providing a picture of the functional connectivity architecture



active during the session. The Spectral Profile summarizes the distribution of EM power across frequency bands — delta, theta, alpha, beta, gamma — for each spatial region covered by the sensor array, giving a frequency-by-region matrix of power values that characterizes the substrate's energetic distribution during the session. Together, the Coherence Topology Map and Spectral Profile constitute the static fingerprint of the session's average regime.

The Cardiac-Neural Coupling Index quantifies the degree to which the cardiac EM rhythm — primarily the R-wave of the QRS complex — entrains or modulates neural regime dynamics. Research has established that cardiac signals influence neural oscillatory dynamics, and the strength of this coupling varies across individuals and states. An elevated coupling index may indicate that cardiac rhythm is exerting unusual influence on neural regime organization, which may be clinically relevant in conditions involving autonomic dysregulation. The Regime Deviation Index places the subject's regime distribution in the context of a normative reference database, expressing as a scalar or vector how far the subject's session deviates from the expected distribution for a demographically matched healthy population.

These outputs together constitute a multidimensional picture of substrate health at the time of acquisition. They are not designed to be reduced to a single diagnostic number — any such reduction would lose the specificity and interpretive richness that makes the framework valuable. They are instead designed to be interpreted alongside the clinical history, self-report, behavioral observation, and structural imaging data that constitute the broader clinical picture. The OmniScan output is a tool for characterizing the substrate state; the clinical interpretation of what that state means for a given individual requires integration with the full context of that individual's presentation.

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## LIMITATIONS OF CURRENT TECHNOLOGY

The first and most immediate limitation is cost and access. OPM-MEG systems capable of whole-head regime-level resolution currently cost in the range of hundreds of thousands of dollars for the sensor array alone, with additional costs for associated electronics, software, and installation. High-performance SQUID-based MEG systems are comparable in cost and require cryogenic maintenance infrastructure. Magnetically shielded rooms, which are necessary for both MEG and high-quality OPM operation, cost several million dollars to design, fabricate, and install. These cost barriers restrict access to a small number of well-funded research institutions globally. High-density EEG is considerably cheaper and more portable, but its spatial resolution and sensitivity to deep sources are substantially inferior to MEG. Until sensor costs fall dramatically — which is plausible as OPM technology matures — regime-level substrate measurement will not be accessible to the majority of clinical settings.

The second limitation is the state of normative databases. Regime classification — assigning a subject's feature vector to a known regime or computing its deviation from a reference — requires a well-characterized reference distribution. Current normative EM substrate databases are small in sample size, demographically narrow (typically drawn from university populations in high-income countries), and not standardized across sites, hardware platforms, or acquisition protocols. A subject scanned with one EEG system in one laboratory cannot reliably be compared against norms derived from a different system in a different laboratory without substantial harmonization effort. Until large, demographically representative, multi-site normative databases exist — with rigorous cross-site validation — regime deviation indices will have limited generalizability.

The third limitation is the inverse problem, described earlier in this paper. Reconstructing the three-dimensional distribution of EM sources inside the brain from measurements on the scalp surface is mathematically underdetermined — there is no unique solution without additional constraints. Current source-reconstruction algorithms address this by incorporating anatomical priors from structural MRI and by applying regularization methods that favor solutions with certain mathematical properties, such as spatial smoothness or minimum norm. These constraints are reasonable but not guaranteed to hold in all individuals, particularly those with atypical anatomy, lesions, or unusual source configurations. The accuracy of source reconstruction — and therefore of the spatial regime maps derived from it — is limited by the validity of these assumptions.

Fourth, OPM sensors are acutely sensitive to motion. Even small head movements during a recording session — a few millimeters of displacement or a fraction of a degree of rotation — can produce field changes at OPM sensors that are comparable in magnitude to the neural signals being measured. Wearable OPM systems that move with the subject's head, reducing this motion sensitivity, are under active development at several research groups, and preliminary results are promising. However, as of the present state of the technology, clinically reliable OPM recordings in environments where subjects cannot maintain near-perfect stillness remain a significant challenge. This limits the applicability of OPM-based regime scanning to cooperative, relatively immobile subjects and makes it difficult to use in pediatric, psychotic, or highly agitated populations.

Fifth, regime classifiers trained on one population or hardware configuration may not generalize to others. A model trained to distinguish healthy from depressive regimes in one dataset, using one sensor type in one shielding environment, may perform poorly when applied to a new dataset acquired under different conditions. The reasons are multiple: different hardware has

different noise floors and frequency responses; different populations may have different baseline regime distributions; and the feature spaces derived from different acquisition setups are not directly comparable without careful normalization. Rigorous cross-site, cross-hardware validation of regime classifiers has not yet been performed at the scale required for clinical confidence.

Sixth, there is the problem of causal ambiguity. Observing that a subject in a depressive episode occupies a particular EM regime, while informative, does not establish whether that regime caused the episode, maintains it, or is simply a parallel correlate of an underlying process driven by other mechanisms. Establishing causal direction requires longitudinal study designs — tracking regime dynamics before, during, and after clinical state changes — and ideally intervention studies where regime perturbation is attempted and the effects on clinical state are measured. These studies are in early stages. Without them, the EM substrate model remains descriptive rather than etiological at the clinical level.

Seventh, there are no established clinical protocols for integrating OmniScan outputs into psychiatric decision-making, and no regulatory frameworks for their use as diagnostic aids in any jurisdiction as of the present time. Clinical adoption of any new diagnostic technology requires not only technical validation but also demonstration of clinical utility — that the information changes clinical decisions in ways that improve patient outcomes — and regulatory review and approval. Neither of these steps has been completed for regime-based substrate scanning. The framework described in this paper currently operates in the research domain, not the clinical one.

Eighth, the long-term reproducibility of an individual's regime map in the absence of clinical change is not yet established. To use regime scanning to track clinical change over time, it is necessary first to know how stable an individual's regime distribution is when their clinical state is stable. If regime

maps vary substantially from session to session due to circadian factors, hydration, caffeine, sleep the night before, or other mundane variables, then detecting clinically meaningful change against this background variability will require either many repeated sessions or statistical methods capable of separating signal from this within-person noise. This baseline characterization of intra-individual regime stability has not been systematically performed in samples large enough to draw general conclusions.

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## **CONCLUSION**

The electromagnetic substrate model offers a physically grounded, measurable way to describe mental states and their disruption. By defining mental health in terms of regime stability and appropriate flexibility within a high-dimensional EM feature space, the framework connects psychiatric phenomenology to a level of description that is, in principle, directly observable with existing and emerging sensor technology. The core ideas — that the brain and body generate structured EM fields, that those fields organize into recognizable and reproducible regime states, and that deviations from adaptive regime dynamics correspond to recognizable psychiatric conditions — are consistent with existing neuroscience and do not require any novel physical claims.

Current technology makes limited versions of this framework viable in research settings. High-density EEG provides accessible regime-relevant data in standard laboratory environments. OPM-MEG systems, while expensive and technically demanding, are beginning to demonstrate the spatial and temporal resolution needed for meaningful regime-level substrate analysis. The pipeline from acquisition to regime classification is well enough

understood to be implemented with current computational and signal-processing tools. What remains missing is not the theoretical framework or the basic technical capability, but the normative databases, cross-site validation studies, longitudinal clinical evidence, and regulatory infrastructure needed to move from research demonstration to clinical utility.

The path forward is clear in outline, if not in timeline. It requires the construction of large, standardized, demographically representative normative substrate databases. It requires longitudinal studies tracking regime dynamics across clinical state changes, with and without treatment. It requires the development and validation of regime classifiers that generalize across hardware platforms and populations. It requires the resolution of the motion-sensitivity and cost barriers that currently limit OPM access. And it requires the development of clinical protocols and regulatory frameworks that specify how regime data should be acquired, interpreted, and acted upon. None of these problems is intractable. Progress on each of them, accumulated over years of coordinated research effort, is what will determine whether the framework described in this paper becomes a useful clinical tool or remains a well-specified theoretical proposal.